

MS101 Makerspace
2024-25/II Spring
Expt 3: DC Power Supply (Ver_Dec31)

Objectives:

- i) To measure and compare the ripple voltages of half and full wave (bridge) rectifiers.
- ii) To compare the output voltage variations and output resistances of the bridge rectifier based DC power supply with that of an IC regulator based DC power supply.

List of components:

Step-down transformer (230 V/12 V), IN4007 diodes, electrolytic capacitors, ceramic disc capacitor, LM 7805, LM 741 and resistors.

Important note:

This experiment is a very basic one, but you need to be extremely careful with your wiring. Incorrect connections of the diodes and/or the electrolytic capacitors can result in explosion and fire. You might severely damage your bread board too. Please note the following:

- i) Electrolytic capacitors used in circuits of Fig.1.3 and Fig. 2.3 have polarity, i.e. the capacitors have +ve and -ve terminals. If the electrolytic capacitor is connected with the wrong polarity, it may explode and may cause fire/injury.
- ii) Similarly, the diodes have anode and cathode terminals, which need to be connected correctly. Wrong connections can result in the diode or the transformer burning. Once again, careless wiring of the diodes may result in injury.
- iii) In order to avoid the above dangerous scenarios, you must get your circuit checked by your TA before switching ON power to the transformer. Also, while making changes, such as changing the C value from 100 μF to 1000 μF , be sure to first switch OFF power to the transformer, and then connect the capacitor observing the correct polarity. Get your connection ok-ed by your TA and then only switch ON the power. It is best to switch OFF power to the transformer while changing the R_L values.

Part A – Unregulated DC Power Supply using Half-wave Rectifier Circuit and a Capacitor Filter

1.1 Half-wave Rectifier

The circuit diagram of a half-wave rectifier circuit is shown below. We will be using a 230 V/12 V step-down transformer and an IN4007 diode (peak forward current: 1A, peak-inverse voltage: 1000 V) to realize the circuit. As shown in the diode image, the cathode lead is identified by the band close to it.

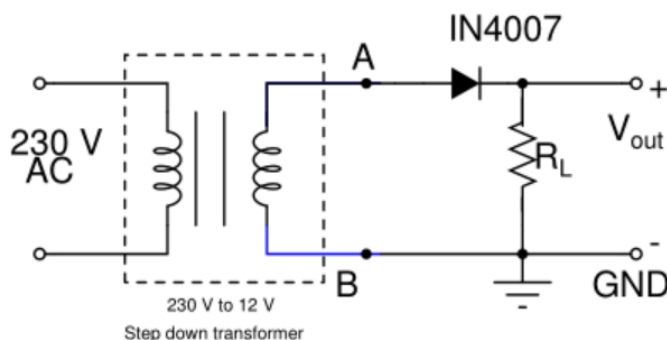


Fig 1.1 Half-wave rectifier circuit

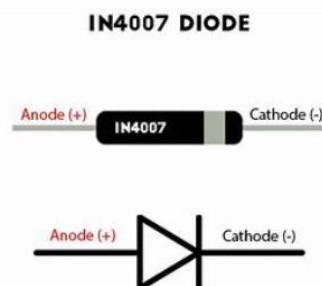


Fig. 1.2 IN4007 diode image

Experiment:

- i) Wire the half-wave rectifier circuit neatly and carefully on the breadboard. $R_L = 1\text{ k}\Omega$. **Take extra care in wiring the diode and the load resistance correctly.**
- ii) Connect CH-1 of the DSO to the transformer output voltage V_A and CH-2 to the rectifier output voltage V_{out} .

Note:

- a) Take extra care to avoid shock. The primary side of the transformer has already been insulated and connected to a plug. You should use only the two leads of the transformer secondary for circuit connections, i.e. points A and B shown in Fig.1.1.
(i) For getting a stable display in CH-1 and CH-2, it is best to use 'AC Line' as the trigger source.
(Use: Trigger Menu > Source > AC Line)

Lab Note Book: Observation and Measurement:

- a) Sketch the V_A and V_{out} waveforms and note down their peak-to-peak voltages.

Note: switch-OFF power to the transformer

1.2 Half-wave Rectifier Circuit with a Capacitor Filter

The half-wave rectifier circuit of Fig. 1.1 is modified and re-drawn in Fig.1.3. In the modified circuit, a large value capacitor ($1000\text{ }\mu\text{F}$) is connected across R_L . Please note from Fig.1.2 the -ve terminal of the capacitor (the -ve terminal is generally the one with the shorter lead length).

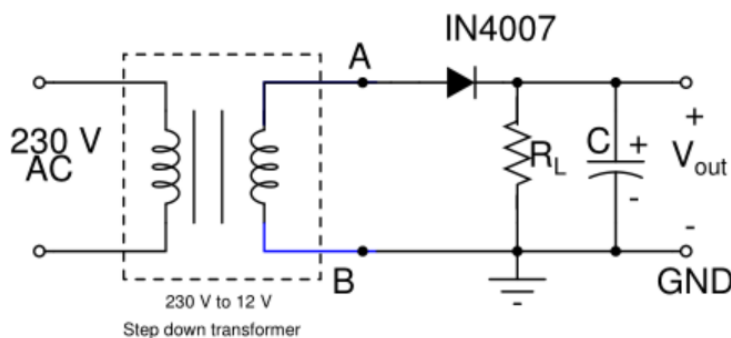


Fig 1.3 Half-wave rectifier circuit with a capacitor filter

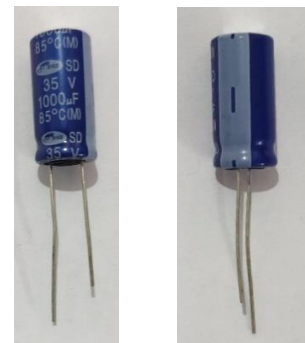


Fig. 1.4 Images of the $1000\text{ }\mu\text{F}$ capacitor

Experiment:

- i) Switch-OFF power to the transformer. Modify the circuit of Fig.1.1 so as to connect the $1000\text{ }\mu\text{F}$ electrolytic capacitor C across R_L . **Take extreme care to see that -ve terminal of the electrolytic capacitor is connected to the GND and the +ve terminal to V_{out} .**
- ii) Switch ON power to the transformer.

Lab Note Book: Observation and Measurement:

- a) Sketch the V_A and V_{out} waveforms for the following values
 - i) $R_L = 1\text{ k}\Omega$, $C = 1000\text{ }\mu\text{F}$.

- For the above case, note down the peak-to-peak ripple voltage and the mean value of V_{out} . (Use the 'Measure' feature of the DSO to obtain the mean value of V_{out}). Please also use your DMM to measure V_{out} . For this measurement the DMM should be in the DC voltage setting.

*Note: Peak-to-peak ripple voltage is relatively much smaller compared to the mean value of V_{out} (say 0.5 V of ripple riding over 16 or 17 V DC). In order to measure the small ripple voltage correctly, you should put the DSO channel to the **AC mode** and choose an appropriate scale.*

Part B – Unregulated DC Power Supply using the Bridge Rectifier Circuit and a Capacitor Filter

2.1 Bridge Rectifier Circuit

Circuit diagram of the bridge rectifier circuit is shown below. We will be using the same 230 V/12 V step-down transformer of Part A. In addition, we will use four IN4007 diodes (peak forward current: 1A, peak-inverse voltage: 1000 V) to realize the circuit. As shown in the diode image (see Fig.2.2), the cathode lead is identified by the band close to it.

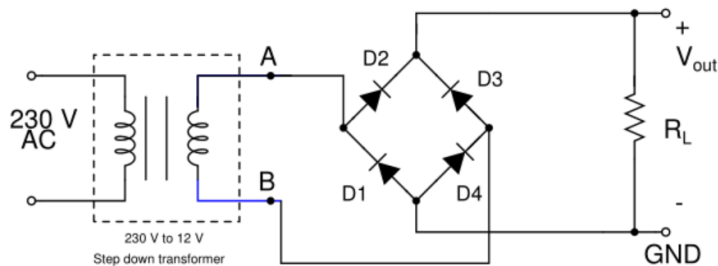


Fig 2.1 Bridge rectifier circuit

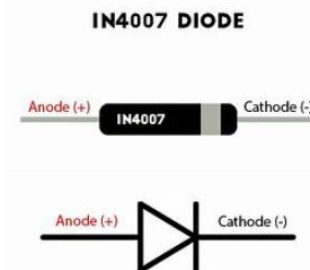


Fig. 2.2 IN4007 diode image

Experiment:

- Wire the bridge rectifier circuit neatly and carefully on the breadboard. $R_L = 10\text{ k}\Omega$. Take extra care in wiring the diodes correctly with the right polarity as shown. **Please note even though the wire from output B of the transformer is crossing the GND line, there is no connection there.**
- Connect DSO (CH-1 or CH-2) only across the rectifier output V_{out} .**
- Note that the GND of the two DSO channels are internally shorted. With the GND connected to the lower terminal of the load, as shown in Fig. 2.1, connecting GND to the transformer secondary outputs (A or B) will short the transformer secondary through one of the diodes. **Hence do not try to display V_{AB} by connecting a DSO probe across the terminals A and B.**

Note:

- For getting a stable display in **CH-1 or CH-2**, it is best to use 'AC Line' as the trigger source. (Use: Trigger Menu > Source > AC Line)

Lab Note Book: Observation and Measurement:

- Sketch the V_{out} waveform (for $R_L = 10\text{ k}\Omega$) and the peak-to-peak voltage.
- After completing (a), switch-OFF power to the transformer.

1.3 Bridge Rectifier Circuit with a Capacitor Filter

The bridge rectifier circuit of Fig.2.1 is modified and re-drawn in Fig.2.3. In the modified circuit, a large value capacitor (100 μF or 1000 μF) is connected across R_L .

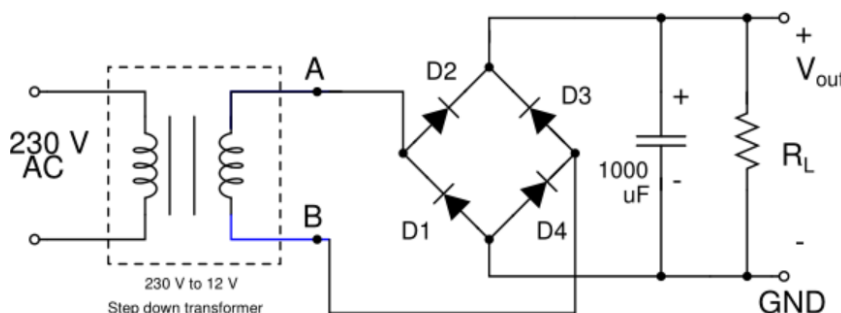


Fig 2.3 Bridge rectifier circuit with a capacitor filter

Experiment:

- i) Switch-OFF power to the transformer. Modify the circuit of Fig. 2.1 so as to connect the $100\ \mu\text{F}$ electrolytic capacitor C across R_L . **Take extreme care to see that –ve terminal of the electrolytic capacitor is connected to the GND and the +ve terminal to V_{out} .**
- ii) Switch ON power to the transformer.

Lab Note Book: Observation and Measurement:

Sketch the V_{out} waveforms for the following values

i) $R_L = 10\ \text{k}\Omega$, $C = 100\ \mu\text{F}$.

ii) $R_L = 3.3\ \text{k}\Omega$, $C = 100\ \mu\text{F}$.

iii) $R_L = 1\ \text{k}\Omega$, $C = 100\ \mu\text{F}$.

- For each of the above cases, note down the peak-to-peak ripple voltage and the mean value of V_{out} ($= V_{\text{out-mean}}$). Use the ‘Measure’ feature of the DSO to obtain $V_{\text{out-mean}}$. Use also your DMM to measure V_{out} . For this measurement the DMM should be in the DC voltage setting.
- Determine the output resistance of the bridge rectifier power supply (with capacitive filter).
Output resistance $= |\Delta V_{\text{out}} / \Delta I_L|$.
 $\Delta V_{\text{out}} = (V_{\text{out-mean}} \text{ for } R_L = 10\ \text{k}\Omega) - (V_{\text{out-mean}} \text{ for } R_L = 1\ \text{k}\Omega)$. ΔI_L is the currents for these cases. For example, I_L (for $R_L = 10\ \text{k}\Omega$) $= V_{\text{out-mean}} / 10\ \text{k}\Omega$. Similarly find I_L for $R_L = 1\ \text{k}\Omega$.

- b) Switch-OFF power to the transformer. **Change C to $1000\ \mu\text{F}$, once again ensuring that –ve of the electrolytic capacitor is connected to the GND and the +ve terminal to V_{out} .**

Switch ON power to the transformer.

Lab Note Book: Observe and sketch the V_{out} waveforms for the following values

i) $R_L = 10\ \text{k}\Omega$, $C = 1000\ \mu\text{F}$.

ii) $R_L = 3.3\ \text{k}\Omega$, $C = 1000\ \mu\text{F}$.

iii) $R_L = 1\ \text{k}\Omega$, $C = 1000\ \mu\text{F}$.

- For each of the above cases, note down the peak-to-peak ripple voltage and the mean value of V_{out} ($= V_{\text{out-mean}}$).
- Determine the output resistance of the above power supply.
Output resistance $= |\Delta V_{\text{out}} / \Delta I_L|$. Use the method as was described earlier to determine ΔV_{out} and ΔI_L .
- Compare the output resistance of the bridge rectifier with $C = 100\ \mu\text{F}$ with the case when $C = 1000\ \mu\text{F}$. Which case is better? Justify your answer.

Part C – IC Regulator (LM7805)

We shall use a very commonly used three-terminal 5V regulator IC to realize a regulated DC power supply. This regulator IC is capable of giving a constant V_{out} ($= 5\text{ V}$) for a large range of V_{in} (for V_{in} varying from 7 V to 25 V). Also, V_{out} will be a constant for a large range of I_L (for I_L varying from 0 to 1 A).

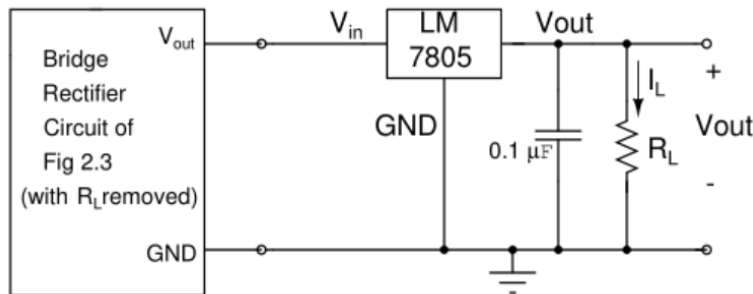


Fig. 3.1 LM7805 regulator circuit



Fig. 3.2 LM7805 IC image

Part C.1 – Measurement of Output Voltage Fluctuations with Load Currents (Output Resistance)

Circuit values: LM7805, $0.1\text{ }\mu\text{F}$; $R_L = 680\text{ }\Omega$, $1\text{ k}\Omega$, and $3.3\text{ k}\Omega$

(The $0.1\text{ }\mu\text{F}$ capacitor is a "ceramic disc capacitor", and it does not have terminal polarities).

For measuring output resistance of the LM7805 IC, we need to vary the load currents while keeping V_{in} to the circuit at a constant value. The output resistance is indicative of the load regulation of LM7805.

Experiment:

1. Wire the circuit of Fig.3.1. **Note that the V_{out} output of the bridge rectifier circuit (of Fig. 2.3, but with R_L removed and $C = 1000\text{ }\mu\text{F}$) is the V_{in} input of the LM 7805 regulator IC.**
2. Identify the V_{in} , GND and V_{out} pins of the LM7805 and take special care to connect the three terminals correctly. The IC terminals may be thicker compared to other components; hence you need to insert the pins carefully. Also, connect the $0.1\text{ }\mu\text{F}$ ceramic capacitor as shown.
3. Vary R_L values: $680\text{ }\Omega$, $1\text{ k}\Omega$, and $3.3\text{ k}\Omega$. For each R_L value, measure V_{out} using your DMM.

Lab Note Book: Observation and Measurement

- a) For noting down your measurement values, tabulate your results into four columns, viz., V_{in} , R_L , V_{out} , I_L (in mA). $I_L = V_{out} / R_L$.
- b) Calculate output resistance of LM7805 as: $|\Delta V_{out} / \Delta I_L|$, where ΔV_{out} is the variation in V_{out} corresponding to the minimum and maximum values of I_L .
- c) Compare the output resistance value you obtained above with that of the bridge rectifier with $C = 1000\text{ }\mu\text{F}$. Which is a better DC supply? Justify your answer.
- d) Refer to the Datasheet of LM7805 IC (see the 'Electrical Characteristics' on page 2) and note the various parameters of the IC. In particular, note down the specified line regulation, load regulation and the output resistance.

Note: Since variations in V_{out} will be very small (in the mV range), your DMM may not be able to indicate these small variations; you will most likely find V_{out} to be a constant for all values of R_L .

Part C.2 – Accurate Measurement of Output Voltage Fluctuations with Load Currents (Output Resistance)

You would have observed that there are hardly any variations in the V_{out} for the given load resistors. This is because of the fact that the V_{out} variations are in the mV range. One way to measure these variations is to use a difference amplifier which measures variations of V_{out} with reference to the nominal V_{out} value.

Circuit values: $+V_{CC} = +12\text{ V}$, $-V_{CC} = -12\text{ V}$; $R_1 = R_3 = 10\text{ k}\Omega$, $R_2 = R_4 = 100\text{ k}\Omega$

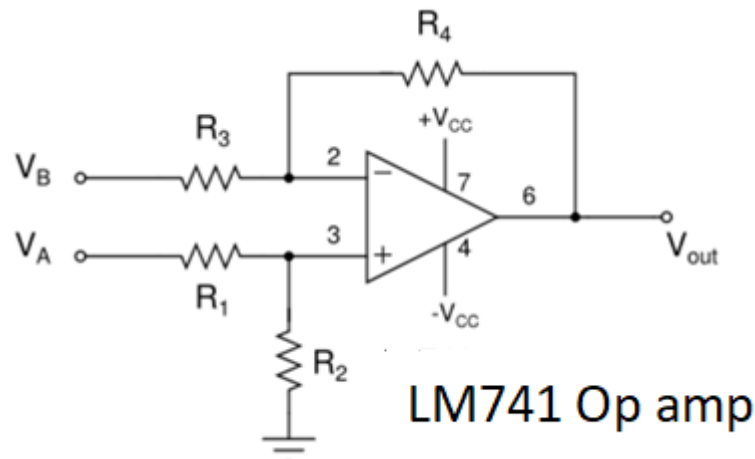


Fig. 3.3 Single-op amp difference amplifier (for measuring V_{out} variations with load)

Fig. 3.3 gives the circuit diagram of a single-op amp difference amplifier. The voltage gain of the amplifier is 10. We want the difference amplifier to measure the small variations in the LM7805 variations with reference to the nominal V_{out} value (of +5V). With $V_A = V_B$, the difference amplifier output will be zero. V_{out} of the difference amplifier will amplify any voltage difference between V_A and V_B by a factor of 10. In this case this will amplify the difference of LM7805 output with reference to its nominal value.

We shall connect V_A input to the LM7805 V_{out} and the V_B input to CH3 of the Keithley Power Supply, set to +5 V. LM7805 V_{out} variations for three R_L values will be measured using the above circuit.

Experiment

1. Wire the op amp difference amplifier circuit of Fig.3.3.
2. Connect V_A input of the difference amplifier to the V_{out} of the LM7805 (Fig.3.1).
3. Using the Ch1 and Ch2 channels of the Keithley Power Supply, set the $+V_{CC}$ and $-V_{CC}$ voltages as +12 V and -12 V respectively for the LM741 op amp.
4. Adjust Ch3 of the Keithley Power Supply to +5V and connect it to V_B input of the difference amplifier.
5. Switch on the Keithley Power Supply.
6. Connect $R_L = 3.3\text{ k}\Omega$. Measure V_{out} of the LM7805 and that of the difference amplifier using your DMM. It should be very close to zero (should be in the range of 10 to 20 mV). The non-zero, but small difference amplifier output voltage may be taken as the error voltage which must be subtracted from all diff amp V_{out} readings.
(In case V_{out} of the difference amplifier is high, measure V_A and V_B values. If they are around +5 V, then the difference amplifier V_{out} will be within 100 mV).
7. For the other two R_L values of LM7805 (1 k Ω and 680 Ω) measure V_{out} of the LM7805 and that of the difference amplifier using your DMM.

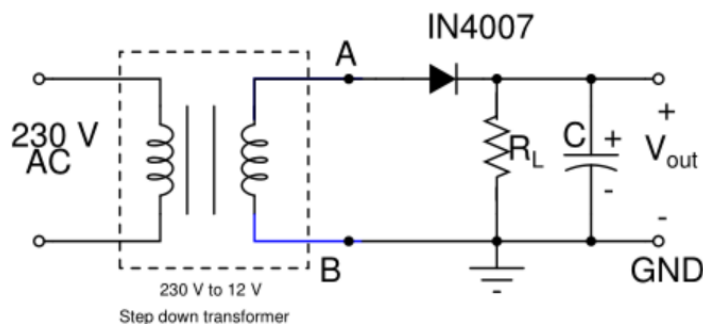
8. Write your results in a Table with the following columns:
 R_L , V_{out} (LM741), I_L (mA), V_{out} (Diff amp) $\Delta V_{out-Diff} = [V_{out} \text{ (Diff amp)}] / 10$
9. Calculate output resistance of LM7805 as: $|\Delta V_{out} / \Delta I_L|$, where ΔV_{out} is the variation in V_{out} corresponding to the minimum and maximum values of I_L .

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Sample Pre-Lab Quiz Questions for Expt 3

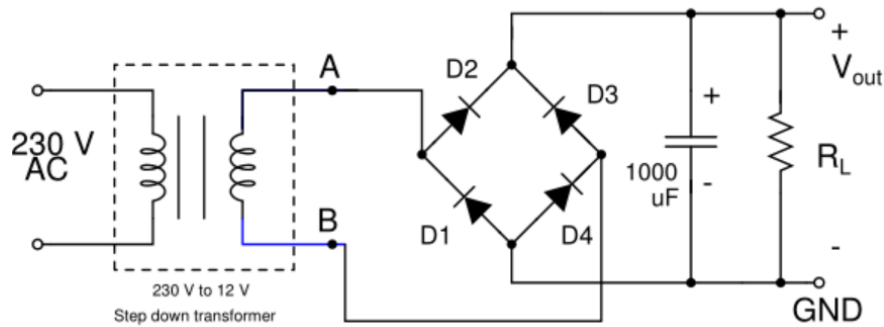
Duration: 10 min; Max marks: 5 marks

1. A rectifier circuit is shown below. A 230 V (RMS) to 12 V (RMS) step down transformer is used as the input to the rectifier circuit. Mark **all the correct options** with regard to the given circuit.



- A) The circuit is a half wave rectifier with capacitor filter.
 - B) The circuit is a full wave rectifier with capacitor filter.
 - C) The diode will conduct only during the positive half cycle of the transformer secondary voltage.
 - D) The diode will conduct only during the negative half cycle of the transformer secondary voltage.
 - E) With the capacitor C not present, V_{out} will be a flat (constant) DC voltage.
 - F) With the capacitor C not present, V_{out} will be a unidirectional, but pulsating DC voltage.
 - G) With the capacitor C not present, V_{out} will be an AC voltage.
-
2. With reference to the rectifier circuit shown above, mark **all the correct options**.
 - A) For a given C (say 100 μF) if R_L is very large (say 1 $\text{M}\Omega$), V_{out} will be almost a flat DC voltage.
 - B) For a given C (say 100 μF) if R_L is reduced, say from 10 $\text{k}\Omega$ to 500 Ω , the ripple voltage in the V_{out} waveform will decrease.
 - C) For a given C (say 100 μF) if R_L is reduced, say from 10 $\text{k}\Omega$ to 500 Ω , the ripple voltage in the V_{out} waveform will increase.
 - D) For a given R_L (say 1 $\text{k}\Omega$) if C is increased, say from 100 μF to 470 μF , the peak-to-peak ripple voltage in the V_{out} waveform will increase.
 - E) For a given R_L (say 1 $\text{k}\Omega$) if C is increased, say from 100 μF to 470 μF , the peak-to-peak ripple voltage in the V_{out} waveform will decrease.

3. A rectifier circuit is shown below. A 230 V (RMS) to 12 V (RMS) step down transformer is used as the input to the rectifier circuit. Mark **all the correct options** with regard to the given circuit.



- A) The circuit is a half wave rectifier with capacitor filter.
 B) The circuit is a full wave rectifier with capacitor filter.
 C) During the positive half cycle of V_{AB} , diodes D3 and D1 will conduct.
 D) During the positive half cycle of V_{AB} , diodes D2 and D4 will conduct.
 E) For the same values of R_L and C , the peak-to-peak ripple voltage in the output voltages (V_{out}) of the circuit in Q.1 and this circuit will be the same.
 F) For the same values of R_L and C , the peak-to-peak ripple voltage in the output voltage (V_{out}) of the circuit in Q.1 will be more than that of this circuit.
 G) For the same values of R_L and C , the peak-to-peak ripple voltage in the output voltage (V_{out}) of the circuit in Q.1 will be less than that of this circuit.
4. With reference to the circuit shown in Q.3, mark **all the correct options**.
 A) The transformer secondary voltage (V_{AB}) should not be observed on the DSO by connecting a DSO probe across A and B.
 B) The transformer secondary voltage (V_{AB}) may be observed on the DSO by connecting a DSO probe across A and B.
 C) The output resistance of a DC power supply made using a full wave rectifier will be more than that made using a half wave rectifier.
 D) The output resistance of a DC power supply made using a full wave rectifier will be less than that made using a half wave rectifier.
 E) The output resistance of a DC power supply made using a full wave rectifier will be the same as that made using a half wave rectifier.
5. Mark **all the correct options** with regard to the LM 7805 regulator DC power supply. The input to the regulator circuit is the output of the bridge rectifier circuit.
 A) For a large range of load currents (say up to 1A), the regulator will give a steady 5 V output.
 B) Only for a medium range of load currents (say up to 100 mA), the regulator will give a steady 5 V output.
 C) Only for a small range of load currents (say up to 20 mA), the regulator will give a steady 5 V output.
 D) For the regulator circuit to work properly, its input voltage should be between 12 V and 25 V.
 E) For the regulator circuit to work properly, its input voltage should be between 15 V and 25 V.
 F) For the regulator circuit to work properly, its input voltage should be between 7 V and 25 V.

---- The End ----